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Effects of cognitive status on disfluency: A cross-linguistic study of Chinese- and English-speaking older adults

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ABSTRACT

Disfluency in spontaneous speech has been reported to be a potential marker for cognitive impairment. However, consistent working definitions are lacking, and it remains unknown whether disfluency measures operate similarly across languages. This study investigates how cognitive status shapes disfluency in Chinese and English, aiming to delineate both language-general patterns and cross-linguistic differences. A total of 100 speech samples elicited via the *Cookie Theft* task (50 Chinese, 50 English) were selected from the MCGD corpus (Chinese) and DementiaBank (English), equally stratified into cognitively normal and impaired groups based on MMSE scores. Following manual annotation, speech samples were analysed using ten disfluency measures across three dimensions: temporal, pausal, and repetition-repair. Results showed that six measures were sensitive to cognitive status across both language groups, and language did not significantly moderate the effect of cognitive impairment on disfluency. Temporal measures declined consistently with cognitive impairment across both languages, with cognitively impaired individuals showing significantly reduced speech rate, phonation-time ratio, and mean length of runs. Pausal measures generally increased with cognitive impairment, with cognitively impaired individuals showing significantly longer silent pauses and more frequent filled pauses. For repetition-repair measures, repetitions increased with cognitive impairment in both languages but showed a higher baseline frequency in Chinese than in English, whereas repairs displayed less consistent patterns across the two language groups. Collectively, these findings support the value of disfluency measures as cross-linguistically relevant markers of cognitive impairment, while underscoring the importance of language-sensitive interpretation for specific disfluency types.

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Introduction

Alzheimer's disease, the most prevalent form of dementia, represents a continuum of cognitive decline, typically progressing from a prodromal stage known as mild cognitive impairment (MCI) to severe dementia (Cummings, 2020; Huang et al., 2023). In contrast to traditional invasive and cost-prohibitive diagnostic approaches, linguistic alterations emerging across this spectrum of cognitive impairment present a more accessible and non-invasive avenue for tracking cognitive deterioration (Yih et al., 2025). Typically, these deficits manifest as

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a decelerated speech rate, increased frequency of pauses and repetitions, and a diminished capacity for syntactic complexity (Arslan & Göksun, 2022; Cheung & Kemper, 1992; Pastoriza-Domínguez et al., 2022; Tucker et al., 2021). These phenomena are intrinsically linked to disfluency, a feature that transcends superficial speech mechanics to reflect underlying cognitive processes (Arslan & Göksun, 2022; Clark, 2002; Fox Tree & Clark, 1997; Kemper et al., 1989; Tucker et al., 2021). Consequently, disfluency has emerged as a candidate indicator of general cognitive impairment, holding profound implications for the non-invasive assessment of language disorders across varying stages of decline (Bortfeld et al., 2001; Fox Tree & Clark, 1997).

Disfluency

Fluency and disfluency are conceptually intertwined, functioning as two poles of a continuum. Characterised by a smooth, rapid, and effortless flow of speech, fluency serves as a cover term for diverse temporal and communicative properties of spoken language (Lennon, 1990). Disfluency, as its counterpart, denotes unintended elements in spontaneous speech that impede the planned flow of expression (Shriberg, 1994). In a seminal classification, Maclay and Osgood (1959) delineated four major categories of disfluency: silent pauses, filled pauses, repetitions, and repairs.

Silent pauses manifest as intervals of silence within continuous speech (Löfgren & Hinzen, 2022; Singh et al., 2001), whereas filled pauses comprise non-lexical items such as *uh* and *um* (de Jong, 2016), alongside lexical fillers such as *I mean* and *you know* (Nasreen et al., 2021). Functionally, these categories exhibit distinct profiles. Silent pauses operate as temporal windows for cognitive processing and self-monitoring during speech formulation, and thus are closely linked to lexical retrieval difficulties, conceptual planning, and syntactic structuring (Chou et al., 2024; Forbes McKay & Venneri, 2005; Löfgren & Hinzen, 2022; Singh et al., 2001). Non-lexical filled pauses, by comparison, primarily serve temporal and interactional functions: they signal planning delays or lexical retrieval difficulties while enabling speakers to retain the conversational floor (Davis & Maclagan, 2010; de Jong, 2016). In contrast, lexical filled pauses underpin discourse-pragmatic organisation, supporting clarification, topic management, and alignment with the listener (Nasreen et al., 2021; Pistono et al., 2024).

According to Shriberg (1994), repetitions involve the immediate recurrence of prior speech segments without alteration to wording or meaning, whereas repairs involve modifying or correcting the original utterance at levels of wording, meaning, or grammar. Functionally, repetitions serve as a compensatory strategy to sustain fluency amidst lexical retrieval difficulties or working memory failures (Mueller et al., 2018; Nicholas et al., 1985). Repairs, conversely, are indicative of speakers' online monitoring and self-regulation processes during speech production (Sajjadi et al., 2012). Specifically, they signal impediments in lexical selection or syntactic formulation, as well as the detection and correction of errors (Fox Tree & Clark, 1997; Levelt, 1983).

Studies of disfluency in older adults

Extending the framework of the four major types of disfluencies introduced above, extensive scholarship on older adults has established that disfluency is intricately linked to cognitive decline. While consistent patterns are observed for certain measures such as

speech rate, findings regarding specific disfluency types remain heterogeneous and sometimes contradictory.

Comparative analyses between younger and older adults suggest that ageing is intrinsically associated with increased disfluency. Relative to their younger counterparts, older adults typically exhibit a slower articulation rate, a higher frequency of silent and filled pauses, and an increased incidence of repetitions and self-repairs (Bóna, 2013; Engelhardt & Markostamou, 2025; Horton et al., 2010; James et al., 2018). Investigations contrasting cognitively impaired older adults with healthy controls further indicate that speech rate and mean length of runs are consistently diminished in impaired groups across diverse linguistic contexts such as English, French, and Chinese. This consistency suggests that these measures may serve as cross-linguistically robust indicators of cognitive decline (Ash et al., 2007; Chou et al., 2024; Forbes McKay & Venneri, 2005; Pistono et al., 2024). Similarly, prolonged silent pauses have been observed across languages. These disruptions are attributed to impairments in semantic memory and executive functioning (Chou et al., 2024; Löfgren & Hinzen, 2022; Pastoriza-Domínguez et al., 2022; Wang et al., 2021).

While the parallel observation of these temporal declines across individual languages is promising, the field lacks direct comparative investigations necessary to validate their cross-linguistic robustness. A specific theoretical concern arises when applying syllable-based disfluency measures, such as speech rate, across diverse linguistic typologies. For example, Chinese and English differ substantially in their syllabic organisation. Chinese syllables conform to a relatively restricted CVX template with a near one-to-one correspondence between syllables and morphemes (Abercrombie, 1967; Duanmu, 2007), whereas English permits complex consonant clusters and extensive vowel reduction, producing a looser mapping between syllabic and lexical units (Ladefoged & Maddieson, 1996). Whether these syllable-based temporal measures retain comparable sensitivity to cognitive decline across languages with such divergent syllabic structures has not been examined.

With respect to filled pauses, repetitions and repairs, findings do not exhibit uniformity. These discrepancies plausibly stem from at least three sources: language-specific properties, task demands, and methodological variations. Regarding linguistic specificity, evidence suggests that English speakers predominantly employ non-lexical filled pauses such as *uh* and *um*, whose frequency escalates significantly among individuals with cognitive impairment (Mueller et al., 2018; Rohanian et al., 2020). In contrast, Chinese speakers demonstrate a greater reliance on lexicalised fillers such as *zhège* ‘this’ and *nàge* ‘that’ to maintain discourse flow; these items constitute more than half of all disfluencies in spontaneous speech (Chou et al., 2024; Tseng, 2006). Collectively, these divergences underscore language-specific disfluency patterns. However, in contrast to the extensive literature on English, research on Chinese, particularly concerning how its language-specific features interact with cognitive decline, remains limited and fragmented.

Beyond language-specific factors, task demands also exert a significant modulatory effect on disfluency patterns. Castro and James (2014) demonstrated that task type profoundly impacts speech production: while younger adults maintained relatively stable disfluency rates across conditions, older adults exhibited increased disfluency when describing negative images versus neutral ones. Furthermore, even when speech is elicited in the same language and via the same task (e.g. the English *Cookie Theft* picture description), variations in analytic methodologies can yield divergent conclusions. For instance, Nicholas et al. (1985) reported significantly higher rates of word repetitions in AD patients relative to

controls using itemised counts of individual repetition. In contrast, Mueller et al. (2016), employing a composite maze ratio that aggregated repetitions, repairs, false starts, and fillers into a single index per utterance, found no significant group difference. Definitions of silent pauses also vary widely, with thresholds ranging from 200 ms (Mueller et al., 2016; Pistono et al., 2019) to 250 ms (Löfgren & Hinzen, 2022) and extending to 2000 ms for long pauses (Chou et al., 2024). Such methodological inconsistencies underscore the imperative for a unified classification system and a computational framework to ensure the comparability of results across studies.

Therefore, several limitations of the current literature exist. First, the absence of consistent terminology and a unified classification system for disfluency limits the comparability and generalisability of existing findings. Definitional heterogeneity is prevalent across studies, with the majority of investigations restricting their scope to a limited subset of disfluency types rather than offering a comprehensive, systematic analysis (Cooper, 1990; Filiou et al., 2020; Kemper et al., 2003). Relatedly, variability in elicitation tasks underscores the need for standardised assessment protocols. Second, despite the existence of isolated studies, the specific relationship between disfluency patterns and cognitive decline in Chinese speakers remains insufficiently characterised. Finally, cross-linguistic and cross-cultural comparisons remain underdeveloped, hindering deeper understanding of how linguistic and cultural contexts shape disfluency and its relevance to cognitive decline.

The present study

To address these issues, the present study examines how cognitive status shapes disfluency patterns in the speech of both Chinese- and English-speaking older adults. To ensure a systematic analysis, we adopted the multidimensional fluency framework proposed by Skehan (2003). Within this theoretical construct, fluency is conceptualised in terms of three dimensions: speed fluency, breakdown fluency, and repair fluency. Specifically, speed or temporal measures include articulation rate and speech rate; breakdown measures encompass the number, duration, and distribution of silent and filled pauses; and repair-related measures cover the frequency of repetitions, reformulations, and self-corrections. Despite its extensive application in second language acquisition research, this framework remains underutilised in investigations of older adults, particularly those with cognitive impairment.

Building on this comprehensive classification, the study employed a hierarchical analytical strategy, progressing from general patterns to granular measures. We first mapped overall trends in disfluency across the three dimensions to characterise how speech changes with cognitive decline, and then narrowed the focus to identify specific measures that effectively distinguish cognitively normal older adults from those with cognitive impairment. Through this approach, the study sought to advance the understanding of disfluency patterns in both Chinese and English and to contribute to the establishment of a more comprehensive system for defining, classifying, and quantifying disfluency. Specifically, we address the following questions:

RQ1: How do speech disfluency measures change with cognitive decline in Chinese- and English-speaking older adults, and are these patterns of change consistent across languages?

RQ2: Which disfluency measures serve as effective indicators of cognitive impairment? Among these, which measures are robust across languages, and to what extent does language moderate these effects?

Methods

Corpora

The Chinese and English speech samples were drawn from the Multimodal Corpus of Gerontic Discourse (MCGD) and DementiaBank, respectively. MCGD is a large-scale, multimodal Chinese corpus developed in China, which collects spontaneous speech from older adults through semi-structured interviews and groups participants according to cognitive status (Zhou, 2024). DementiaBank, which is part of the TalkBank platform, provides speech samples from older adults with and without dementia (Lanzi et al., 2023). For the present study, the speech samples in both languages were elicited using the *Cookie Theft* picture description task from the *Boston Diagnostic Aphasia Examination* (Goodglass & Kaplan, 1983).

Participants were classified into a cognitively normal (CN) group and a cognitively impaired (CI) group based on their Mini-Mental State Examination (MMSE) scores. In total, 100 speech samples were analysed, comprising 50 Chinese (25 CN, 25 CI) and 50 English (25 CN, 25 CI). We used the widely accepted cut-off points proposed by Tombaugh and McIntyre (1992), pp. 24–30 for no cognitive impairment, 18–23 for mild cognitive impairment (MCI), and 0–17 for severe cognitive impairment (SCI). In the Chinese dataset, participants in the CI group had MMSE scores ranging from 10 to 23, comprising 13 individuals with MCI and 12 with SCI. Similarly, the English CI group exhibited a comparable MMSE range (10–23), consisting of 10 participants with MCI and 15 with SCI. Given the practical constraints on clinical data availability, further stratification by severity was not feasible within the present sample.

Statistical analyses were conducted separately for each language dataset to verify demographic matching between the CN and CI groups. Age differences were assessed using independent t-tests, and MMSE scores were analysed using the Mann-Whitney U test. Sex distribution was examined using Fisher's Exact Test for the English group due to small expected cell counts, whereas the Chi-square test was applied to the Chinese group. As shown in Table 1, only MMSE scores differed significantly between the CN and CI groups in both datasets, confirming that the groups were well-matched in terms of age and sex.

Measures

Guided by Skehan's (2003) framework and measures in the literature on older adults and disfluency, we selected a core set of measures and stratified them into temporal, pausal, and repetition-repair measures. To ensure methodological rigour, working definitions and computational units were standardised across all measures (see Table 2).

Temporal measures evaluate the temporal dynamics of speech production, focusing on the speed and continuity of production. This category includes the following four measures:

Table 1. Demographic data of participants.

Group	Variable	CN (N = 25)	CI (N = 25)	p
English	Age (M ±SD)	69.32 ± 5.65	70.36 ± 7.33	0.5769
	MMSE (M ±SD)	29.48 ± 0.65	16.68 ± 3.31	0.0000*
	Sex (M/F)	17 / 8	21 / 4	0.3209
Chinese	Age (M ±SD)	70.04 ± 6.18	72.52 ± 7.49	0.2080
	MMSE (M ±SD)	27.84 ± 1.28	17.92 ± 3.73	0.0000*
	Sex (M/F)	11 / 14	7 / 18	0.3768

Note: M: mean, SD: standard deviation, CN: cognitively normal, CI: cognitively impaired.

Table 2. Disfluency measures.

Category	Measure	Abbreviations	Calculation Method	References
Temporal Measures	Speech rate	SR	Number of syllables / Speaking time (including pauses)	Towell et al. (1996), Gayraud et al. (2011), Bóna (2014)
	Articulation rate	AR	Number of syllables / Phonation time (excluding silent pauses)	
	Phonation-time ratio	PTR	(Phonation time / Speaking time) × 100	Towell et al. (1996)
	Mean length of runs	MLR	Syllables / Number of significant pauses	
Pausal Measures	Mean length of silent pauses	MLSP	(Duration of SP × 1000) / Number of SP	Löfgren and Hinzen (2022)
	Mean length of filled pauses	MLFP	(Duration of FP × 1000) / Number of FP	
	Frequency of silent pauses	FSP	(Number of SP × 100) / Words	Bóna (2014), Pistono et al. (2019)
	Frequency of filled pauses	FFP	(Number of FP × 100) / Words	
Repetition-Repair Measures	Frequency of repetitions	FR	(Number of repetitions × 100) / Words	Pistono et al. (2024)
	Frequency of repairs	FRE	(Number of repairs × 100) / Words	

- (1) Speech rate (SR): defined as the number of syllables produced per second of total duration, including silent intervals.
- (2) Articulation rate (AR): measures the number of syllables per second of phonation time (excluding silent pauses), thus indexing the speed of actual articulatory execution.
- (3) Phonation-time ratio (PTR): quantifies the proportion of actual phonation time relative to the total speaking time.
- (4) Mean length of runs (MLR): calculated as the average number of syllables produced between significant pauses (exceeding 200 ms).

Pausal measures capture the frequency and mean length of both silent (FSP and MLSP) and filled pauses (FFP and MLFP), signalling interruptions in the flow of speech planning. Concerning perceptible pauses, while they typically range from 200 ms to 250 ms (Zellner, 1994), a threshold of 200 ms was adopted in this study to align with previous research on Alzheimer’s disease (Gayraud et al., 2011; Löfgren & Hinzen, 2022; Pistono et al., 2019). Regarding filled pauses, in addition to typical items such as *um* and *uh*, specific Chinese lexical items were classified based on context. Consistent with Chou et al. (2024) and Xie et al. (2023), words such as *zhège* ‘this’ and *nàge* ‘that’ can be viewed as filled pauses when they are devoid of referential meaning and function primarily as time-buying devices rather

than demonstratives. Analogously, words like *duiba* ‘right’ in Chinese can be treated as lexical fillers because they serve the same semantic function as fillers like *you know* in English.

Finally, we examined repetition-repair measures. Although Skehan originally designated this dimension simply as ‘repair’, we refined the terminology to ‘repetition-repair’ to explicitly distinguish between repetition (repeating content without change) and repair (modifying content). This category quantifies the frequency of repetitions (FR) and repairs (FRE) in spontaneous speech. Crucially, in Chinese, when the non-referential forms of *zhège* ‘this’ and *nàge* ‘that’ occurred repeatedly in immediate succession, all occurrences in the sequence were coded as filled pauses, while the second and subsequent occurrences were additionally counted as repetitions because they constituted immediate repetitions of the preceding filler item.

Data processing

A total of 100 speech samples were analysed, including 50 Chinese samples (25 from the cognitively normal group and 25 from the cognitively impaired group) and 50 English samples (also equally divided). After data collection, the recordings were processed and manually annotated using Praat software (Boersma & Weenink, 2024). To operationalise the identification of disfluency phenomena, each file was transcribed and annotated on multiple tiers: sentence (transcribed units segmented sentence-by-sentence), pause_silent (silent pauses), pause_filled (filled pauses), repeat (repetitions), repair (repairs), syllable (syllable counts), and T_total (total speech duration, measured from the onset of the first word to the offset of the last word, including all internal silent pauses). Following annotation, all data were processed through Python scripts for feature extraction and visualisation to address the first research question. To address the second research question, all disfluency measures were analysed using two-way permutation ANOVA (5000 iterations) with language and group as between-subject factors. This method is distribution-free and does not assume normality or homogeneity of variance, making it appropriate for the present data. To control for multiple comparisons, the Benjamini-Hochberg false discovery rate (FDR) correction was applied within each effect type separately. Both raw and FDR-corrected *p*-values are reported, with FDR-corrected $p < 0.05$ considered statistically significant.

All TextGrid manipulations were executed via the praatio library (Mahrt, 2023). Data processing and visualisation were conducted in Python using NumPy (Harris et al., 2020), pandas (The pandas development team, 2023), and Matplotlib (Hunter, 2007). Permutation ANOVA was performed in R using the lmPerm package (Wheeler & Torchiano, 2016).

Results

General changes of measures

To facilitate comparative visualisation across measures, all raw values were standardised to T-scores (see Figures 1–3) and the original data are presented in Table 3.

Regarding the trajectory in disfluency measures associated with cognitive decline, temporal measures demonstrated a consistent decrement from the CN to the CI groups in both languages (see Figure 1). Notably, mean length of runs displayed greater individual variability in the Chinese CN group compared to the CI group, a pattern not observed in the

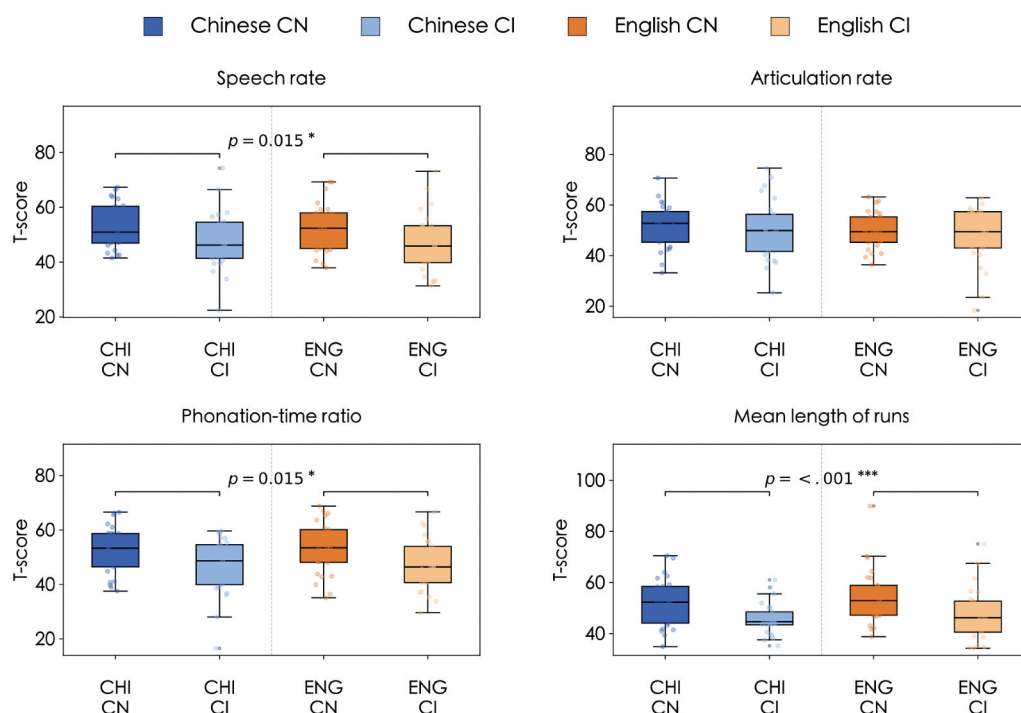


Figure 1. Boxplots of the four temporal measures. To enable cross-measure comparison, all values were converted to T-scores. Significant differences are marked with asterisks (** $p_{FDR} < 0.001$; * $p_{FDR} < 0.05$). Original units: speech rate (syllables/s); Articulation rate (syllables/s); phonation-time ratio (%); mean length of runs (syllables/run). Abbreviations: CN = cognitively normal; CI = cognitively impaired.

English cohort. Meanwhile, articulation rate remained relatively stable across cognitive status in the English group.

Pausal measures displayed a general upward trend associated with cognitive impairment in both language groups (see Figure 2). However, the data distributions revealed considerable instability, characterised by a prevalence of outliers and elevated variability, highlighting the complex nature of pausing behaviour. Specifically, the frequency of silent pauses demonstrated marked inter-individual variability within the English CN group.

Repetition-repair measures showed divergent patterns (see Figure 3). While the frequency of repetitions escalated in both language groups, repairs exhibited opposing trends. Furthermore, the English CN group presented with numerous outliers for repetitions, indicative of a zero-inflated distribution wherein the majority of speakers produced no repetitions.

Sensitive measures of cognitive impairment

Results of statistical comparisons across groups and languages are presented in Table 4. Among the ten measures analysed, the two-way permutation ANOVAs yielded significant main effects of group (cognitive status) for six measures: speech rate ($F(1, 96) = 7.55$, $p =$

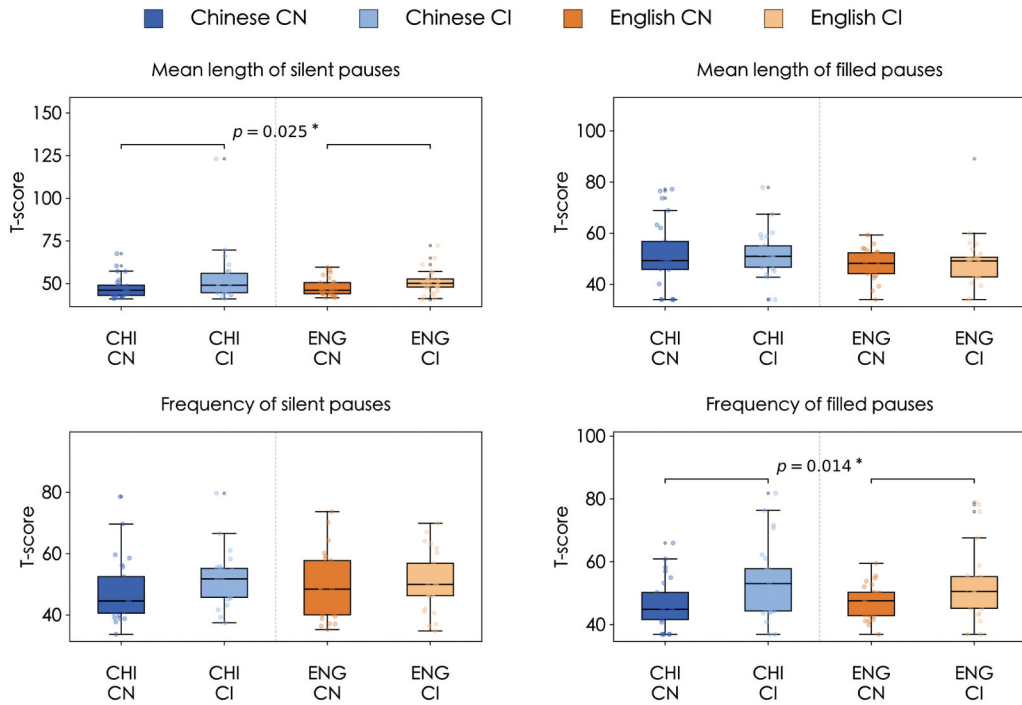


Figure 2. Boxplots of the four pausal measures. To enable cross-measure comparison, all values were converted to T-scores. Significant differences are marked with asterisks (* $p_{FDR} < 0.05$). Original units: mean length of silent pauses (ms); mean length of filled pauses (ms); frequency of silent pauses (per 100 words); frequency of filled pauses (per 100 words). Abbreviations: CN = cognitively normal; CI = cognitively impaired.

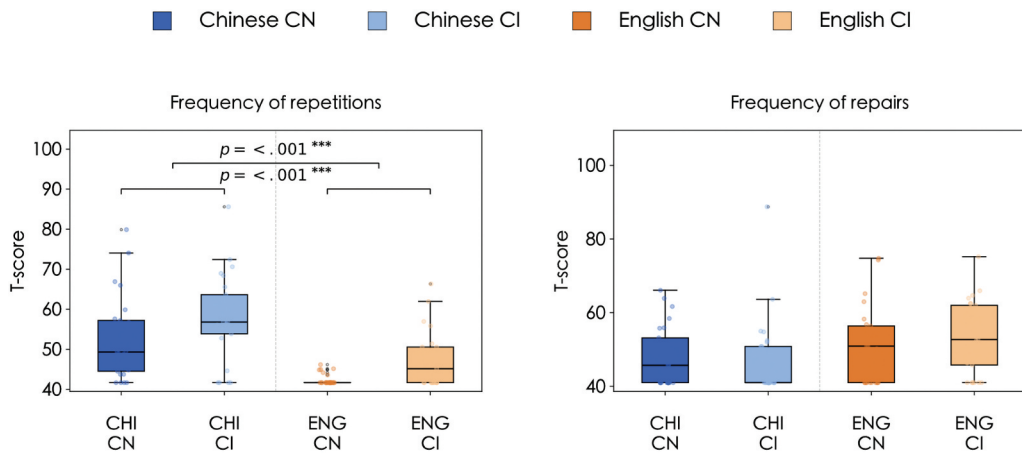


Figure 3. Boxplots of the two repetition-repair measures. To enable cross-measure comparison, all values were converted to T-scores. Significant differences are marked with asterisks (***) $p_{FDR} < 0.001$. Original units: frequency of repetitions (per 100 words); frequency of repairs (per 100 words). Abbreviations: CN = cognitively normal; CI = cognitively impaired.

Table 3. Descriptive statistics for all disfluency measures by language and group.

Measure	Chinese CN	Chinese CI	English CN	English CI
SR (M $\pm$ SD)	2.51 $\pm$ 0.59	2.13 $\pm$ 0.77	2.50 $\pm$ 0.64	2.10 $\pm$ 0.78
AR (M $\pm$ SD)	4.14 $\pm$ 0.61	4.03 $\pm$ 0.81	3.99 $\pm$ 0.50	3.84 $\pm$ 0.78
PTR (M $\pm$ SD)	60.97 $\pm$ 12.71	52.70 $\pm$ 15.07	62.58 $\pm$ 13.78	54.05 $\pm$ 13.57
MLR (M $\pm$ SD)	7.89 $\pm$ 2.54	6.08 $\pm$ 1.69	8.40 $\pm$ 3.03	6.52 $\pm$ 2.79
MLSP (M $\pm$ SD)	1526 $\pm$ 828	2218 $\pm$ 2037	1517 $\pm$ 659	1914 $\pm$ 907
MLFP (M $\pm$ SD)	485 $\pm$ 375	503 $\pm$ 267	399 $\pm$ 173	435 $\pm$ 298
FSP (M $\pm$ SD)	11.79 $\pm$ 4.38	13.39 $\pm$ 3.71	12.39 $\pm$ 4.54	13.02 $\pm$ 4.03
FFP (M $\pm$ SD)	2.65 $\pm$ 2.25	4.68 $\pm$ 3.37	2.95 $\pm$ 1.57	4.44 $\pm$ 3.25
FR (M $\pm$ SD)	2.86 $\pm$ 2.72	4.23 $\pm$ 2.83	0.20 $\pm$ 0.37	1.43 $\pm$ 1.80
FRE (M $\pm$ SD)	0.74 $\pm$ 0.90	0.70 $\pm$ 1.16	1.10 $\pm$ 1.13	1.36 $\pm$ 1.06

Note: M = mean; SD = standard deviation; CN = cognitively normal; CI = cognitively impaired; SR = Speech rate (syllables/s); AR = Articulation rate (syllables/s); PTR = Phonation-time ratio (%); MLR = Mean length of runs (syllables/run); MLSP = Mean length of silent pauses (ms); MLFP = Mean length of filled pauses (ms); FSP = Frequency of silent pauses (per 100 words); FFP = Frequency of filled pauses (per 100 words); FR = Frequency of repetitions (per 100 words); FRE = Frequency of repairs (per 100 words).

Table 4. Permutation ANOVA results for all disfluency measures.

Measure	Effect	$F(1, 96)$	p (perm)	p (FDR)
Speech rate	Language	0.018	1.000	1.000
Speech rate	Group	7.549	0.007	0.015 *
Speech rate	Language $\times$ Group	0.002	0.902	1.000
Articulation rate	Language	1.575	0.168	0.420
Articulation rate	Group	0.875	0.322	0.358
Articulation rate	Language $\times$ Group	0.021	0.941	1.000
Phonation-time ratio	Language	0.286	0.603	1.000
Phonation-time ratio	Group	9.249	0.007	0.015 *
Phonation-time ratio	Language $\times$ Group	0.002	0.824	1.000
Mean length of runs	Language	0.856	1.000	1.000
Mean length of runs	Group	12.995	<0.001	<0.001 ***
Mean length of runs	Language $\times$ Group	0.005	1.000	1.000
Mean length of silent pauses	Language	0.401	0.581	1.000
Mean length of silent pauses	Group	4.857	0.015	0.025 *
Mean length of silent pauses	Language $\times$ Group	0.358	0.603	1.000
Mean length of filled pauses	Language	1.799	0.099	0.330
Mean length of filled pauses	Group	0.211	0.804	0.804
Mean length of filled pauses	Language $\times$ Group	0.025	0.902	1.000
Frequency of silent pauses	Language	0.018	0.980	1.000
Frequency of silent pauses	Group	1.773	0.279	0.358
Frequency of silent pauses	Language $\times$ Group	0.338	1.000	1.000
Frequency of filled pauses	Language	0.003	1.000	1.000
Frequency of filled pauses	Group	10.528	0.004	0.014 *
Frequency of filled pauses	Language $\times$ Group	0.242	0.258	1.000
Frequency of repetitions	Language	39.751	<0.001	<0.001 ***
Frequency of repetitions	Group	9.002	<0.001	<0.001 ***
Frequency of repetitions	Language $\times$ Group	0.025	1.000	1.000
Frequency of repairs	Language	5.728	0.011	0.053
Frequency of repairs	Group	0.283	0.322	0.358
Frequency of repairs	Language $\times$ Group	0.513	0.462	1.000

Note: p (perm) = permutation-based p -value. p (FDR) = Benjamini-Hochberg FDR corrected within each effect type. *** p_{FDR} < 0.001; ** p_{FDR} < 0.01; * p_{FDR} < 0.05.

0.015), phonation-time ratio ($F(1, 96) = 9.25$, $p = 0.015$), mean length of runs ($F(1, 96) = 13.00$, $p < 0.001$), mean length of silent pauses ($F(1, 96) = 4.86$, $p = 0.025$), frequency of filled pauses ($F(1, 96) = 10.53$, $p = 0.014$), and frequency of repetitions ($F(1, 96) = 9.00$, $p < 0.001$). Compared to cognitively normal controls, participants with cognitive impairment exhibited significantly lower speech rate, phonation-time ratio, and mean length of runs, and

significantly higher mean length of silent pauses, frequency of filled pauses, and frequency of repetitions. Additionally, a significant main effect of language was found for frequency of repetitions ($F(1, 96) = 39.75, p < 0.001$), driven by higher baseline rates among Chinese speakers. Crucially, no language $\times$ group interaction terms survived FDR correction for any of the measures.

Discussion

The present study examined disfluency measures associated with cognitive decline in Chinese- and English-speaking older adults. Adopting a hierarchical analytical strategy, we progressed from characterising general trends to isolating sensitive measures.

General changes of measures

Temporal measures

Regarding temporal measures (see [Figure 1](#)), speech rate (SR), phonation-time ratio (PTR), and mean length of runs (MLR) demonstrated stable cross-linguistic patterns of decline associated with cognitive impairment. Both language groups showed a uniform decrement in these three measures in the CI groups, indicating that reduced cognitive ability is broadly associated with diminished fluency and fragmented speech segments. Notably, the Chinese CN group exhibited pronounced individual variability in MLR, suggesting that this measure may be susceptible to sociodemographic factors such as educational background or idiosyncratic speech habits.

In contrast, articulation rate (AR) showed a stable pattern across groups and languages. Because AR is calculated by excluding all silent pause durations, it captures the speed of articulation by filtering out the cognitive time spent on lexical retrieval and syntactic planning. Its stability across cognitive groups is consistent with prior evidence from French-speaking populations (Gayraud et al., 2011) which reported AR did not significantly differ between AD patients and healthy controls. The cross-linguistic stability of AR suggests that the motor aspects of speech production are relatively preserved in early-stage cognitive impairment (Forbes McKay & Venneri, 2005; Gayraud et al., 2011).

Pausal measures

Pausal measures generally trended upward in the CI groups across both languages (see [Figure 2](#)), reflecting heightened hesitation, prolonged latency for lexical retrieval, or the strategic use of pauses to maintain the conversational floor. However, despite these shared trends, the data distributions revealed substantial heterogeneity. First, the English CN group displayed pronounced inter-individual variability in the frequency of silent pauses. This appears to be driven by the nature of the elicitation task. The *Cookie Theft* stimulus depicts a domestic kitchen scene deeply rooted in Western culture, thereby resonating strongly with native English speakers. Such familiarity likely prompted some English participants to attempt more elaborate descriptions. Consequently, this increased cognitive effort to provide comprehensive details can lead to more frequent silent pauses, particularly in the later stages of the narrative.

Second, methodological discrepancies in corpus construction may have confounded the observed trends. In the Chinese dataset, interviewers frequently employed backchannel

signals (e.g. *en* ‘yes’) to acknowledge participants’ responses. This interaction style may have inadvertently inflated both the duration and frequency of silent pauses, thereby introducing methodological noise that impedes precise cross-linguistic comparison.

Consequently, while silent and filled pauses hold promise as indicators of cognitive decline, their interpretability is constrained by their inherent sensitivity to external variables. This observation underscores that, beyond refining definitions, establishing unified protocols for corpus construction and implementing broad, culturally adapted task types are critical foundations for accurately elucidating the mechanisms underlying pausal behaviour.

Repetition-repair measures

Concerning repetition-repair measures, both language groups exhibited an overall increase in frequency of repetitions (FR) among individuals with cognitive impairment, potentially reflecting disrupted lexical retrieval or syntactic planning. In the Chinese group, recurrent employment of *zhège* ‘this’ and *nàge* ‘that’ among participants in the CI group was particularly salient, especially immediately preceding nouns. This pattern indicates that these specific items serve as critical indicators signalling lexical retrieval difficulties, or more precisely, challenges in object recognition. Conversely, the English group displayed a more heterogeneous pattern of repetition. The forms of repetition were markedly variable, spanning multiple grammatical categories including both content words (e.g. nouns) and function words (e.g. conjunctions).

Divergent tendencies in repair behaviours emerged between the two languages: the CN group demonstrated a higher frequency of repairs in Chinese, whereas the CI group exhibited this pattern in English. A qualitative examination of the annotated data revealed that repairs operated at distinct linguistic levels across the two languages. In Chinese, repairs predominantly targeted lexical retrieval, specifically object naming, whereas in English, they more frequently involved the restructuring of sentence-level units. Interpreting these divergent patterns requires acknowledging that repair is a complex disfluency dependent on two successive prerequisites: the generation of an error and its subsequent detection and correction via self-monitoring. Cognitive decline impacts these prerequisites in opposing directions. On one hand, cognitive impairment increases the frequency of lexical and syntactic errors, thereby expanding the pool of potential repair targets (Sajjadi et al., 2012). On the other hand, it degrades the self-monitoring capacity required to detect and act upon these errors (Levelt, 1983), reducing the likelihood of successful execution. The repair frequency thus reflects the interplay of these countervailing forces, necessitating a cautious interpretation of the current findings. Furthermore, methodological limitations must be considered. The present study relied on a single elicitation task; although the *Cookie Theft* picture ensures structural comparability across languages, cultural familiarity with the depicted scene differed between the groups. This asymmetry may have disproportionately influenced both the volume of speech produced and the specific types of repairs initiated. Future studies employing a broader range of tasks and culturally balanced stimuli are essential to disentangle task-driven repair behaviours from potential cognition- or language-specific monitoring patterns.

Sensitive measures of cognitive impairment

As presented in Table 4, the two-way permutation ANOVAs revealed significant main effects of group for six out of the ten disfluency measures: speech rate (SR), phonation-time ratio (PTR),

mean length of runs (MLR), mean length of silent pauses (MLSP), frequency of filled pauses (FFP), and frequency of repetitions (FR). Regarding language differences, a significant main effect of language was observed for FR. Frequency of repairs (FRE) showed only a marginal main effect of language, which warrants cautious interpretation. Critically, no language $\times$ group interaction terms survived FDR correction for any of the ten measures. This indicates that language did not significantly moderate the effect of cognitive status on any disfluency measure.

Temporal measures

Regarding temporal measures, the results show that in both Chinese and English groups, SR, PTR and MLR consistently decline in the CI group. These temporal disruptions were robust across both Chinese and English speakers, with no significant language $\times$ group interaction. The findings are consistent with prior research that has documented significant declines in these temporal markers within individual languages, including English (Ash et al., 2007), French (Gayraud et al., 2011), Spanish (Pastoriza-Domínguez et al., 2022), and Mandarin Chinese (Chou et al., 2024).

SR, PTR and MLR collectively quantify the continuous flow and temporal organisation of discourse. A concurrent decline in these measures manifests as a highly fragmented speech stream characterised by shorter continuous utterances and a higher density of silent moments. Cognitively, this degradation is driven by two intertwined mechanisms. First, the reduction in MLR reflects a decline in working memory capacity and executive functioning, which restricts the patients' ability to maintain long-distance syntactic pre-planning (Ash et al., 2007). Second, the progressive impairment of semantic networks leads to profound lexical retrieval difficulties. To compensate for the increased cognitive load during word-finding, patients are forced to insert silent pauses as online planning buffers, which reduces the proportion of phonation time and slows down the global speech rate (Forbes McKay & Venneri, 2005; Pastoriza-Domínguez et al., 2022).

While prior research has documented significant declines in these measures within individual languages, the current study extends these findings by demonstrating that this sensitivity is preserved across two languages with fundamentally different syllabic organisations, reinforcing their robustness as language-general markers.

Pausal measures

With respect to pausal measures, among the four measures analysed, mean length of silent pauses (MLSP) and frequency of filled pauses (FFP) emerged as significant measures differentiating the CI from the CN group, with no significant language $\times$ group interactions detected. In contrast, mean length of filled pauses (MLFP) and frequency of silent pauses (FSP) did not differ significantly between groups.

The finding that MLSP is elevated in cognitively impaired individuals converges with evidence from Spanish (Pastoriza-Domínguez et al., 2022), French (Pistono et al., 2019), and Chinese (Chou et al., 2024), suggesting that prolonged silent pause duration constitutes a language-general marker of cognitive decline. The significant group effect for FFP, however, appears inconsistent with Chou et al. (2024) and Löfgren and Hinzen (2022), both of whom reported no significant filler differences between groups. A plausible explanation lies in the definitional scope of filled pauses. These prior studies restricted their categories to non-lexical items such as *uh* and *um*, whereas the present study included both non-lexical and lexical filled pauses across both languages. While English incorporates

lexical fillers such as *well* or *you know*, in Chinese, fillers such as *zhège* ‘this’ and *nàge* ‘that’ are particularly prominent and frequently appear in repeated sequences before lexical targets (Chou et al., 2024). The tendency for this repetition-filler hybrid to precede nouns suggests that older adults with CI recruit these forms to manage cognitive load during lexical retrieval (Xie et al., 2023).

However, one critical methodological limitation of the present study is the conflation of lexical and non-lexical filled pauses into a single overarching category. While this inclusive criterion enhanced the overall capture of pausal behaviours across languages, it inherently sacrifices analytical granularity. Lexical and non-lexical fillers may serve distinct pragmatic functions or reflect breakdowns at different stages of lexical retrieval. By conflating these subtypes, the current analysis precludes the investigation of whether cognitive impairment disproportionately disrupts specific filler modalities.

Repetition-repair measures

With respect to repetition-repair measures, among the two measures analysed, frequency of repetitions (FR) exhibited significant main effects of both language and group, with no significant language $\times$ group interaction detected. Frequency of repairs (FRE) did not differ significantly between groups and languages.

The significant group effect for FR indicates that CI participants produced more repetitions than their cognitively normal counterparts, a finding consistent with Nicholas et al. (1985), who likewise reported elevated repetition rates in AD patients during *Cookie Theft* descriptions. Functionally, these repetitions operate as a syntactic planning buffer, allowing speakers to hold the floor while organising upcoming speech (Tseng, 2006). The significant language effect, with Chinese speakers producing markedly more repetitions than English speakers regardless of cognitive status, points to a baseline disparity between the two languages. Importantly, the absence of a significant language $\times$ group interaction indicates that, despite this baseline disparity, the effect of cognitive impairment on repetitions was consistent across the two languages. Such a disparity may originate from the repeated occurrence of non-referential forms such as *zhège* ‘this’ and *nàge* ‘that’ before lexical targets during word retrieval. Although these forms may function analogously to English fillers such as *um* and *uh* by serving as buffers for lexical retrieval, their surface form differs from English non-lexical fillers. How to accurately measure these compound disfluencies without biasing cross-linguistic comparisons remains an important methodological challenge for future studies.

Limitations

Several limitations of the present study warrant consideration. First, the cognitively impaired group included participants with mild and severe cognitive impairment due to sample size constraints within the corpora. This clinical heterogeneity restricts the current findings to characterising cognitive impairment broadly, precluding determinations of whether specific disfluency measures are differentially sensitive to early versus advanced stages of decline. Second, the reliance on a single elicitation task introduces potential confounds. Although the *Cookie Theft* picture ensures structural comparability across languages, varying degrees of cultural familiarity with the depicted scene may have disproportionately influenced both speech volume and the specific types of disfluencies produced. Third, the study operationalised disfluency measures using broad formal categories. As demonstrated by the

divergent patterns observed for repetitions and repairs, a single formal label can subsume functionally heterogeneous phenomena in cross-linguistic comparisons. For instance, the Chinese demonstratives *zhège* ‘this’ and *nàge* ‘that’ can function as lexical filled pauses, while repeated sequences of these forms can additionally contribute to repetition measures. This illustrates how a single surface form can contribute to multiple disfluency measures depending on its discourse function and local sequential structure. Future research must address these limitations by employing finer-grained, functionally differentiated annotation schemes across diverse elicitation tasks and larger, severity-stratified clinical cohorts.

Conclusions

The present study examined how disfluency measures vary with cognitive status among older adults with different language backgrounds. The results demonstrate that cognitive decline is associated with systematic alterations in multiple disfluency measures, though the direction and magnitude of these changes exhibit cross-linguistic variability. Notably, temporal measures displayed relatively unified trends across cognitive states in both languages, while pausal measures manifested considerable instability. Among the ten measures examined, six demonstrated significant sensitivity to cognitive status. Speech rate, phonation-time ratio, and mean length of runs were significantly lower in the CI group, whereas mean length of silent pauses, frequency of filled pauses, and frequency of repetitions were significantly higher. Critically, the impact of cognitive impairment on these disfluency measures is broadly consistent across Chinese and English, despite the substantial typological distance between the two languages, suggesting that these six measures reflect domain-general cognitive processes, such as working memory and executive control. Beyond these cognitive effects, the frequency of repetitions also showed a significant main effect of language, with Chinese speakers producing markedly more repetitions than English speakers across both cognitive groups. This baseline disparity suggests that the distribution of repetition phenomena may differ across languages, even though the effect of cognitive impairment on repetitions was broadly consistent across Chinese and English. Collectively, these findings advance our understanding of how spontaneous speech indexes cognitive functioning in ageing populations, highlighting the critical value of cross-linguistic perspectives in refining disfluency-based measures of cognitive impairment.

Author contributions

CRediT: **Tongfu Yang:** Data curation, Methodology, Visualization, Writing – original draft; **Lihe Huang:** Conceptualization, Investigation, Project administration, Resources, Software, Supervision, Validation, Writing – review & editing; **Tsy Yih:** Conceptualization, Funding acquisition, Project administration, Supervision, Validation, Writing – review & editing.

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